

Solution Architecture

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Project Name	Import Data Using Transform Maps in Sheets
Maximum Marks	4 Marks

Solution Architecture

Solution architecture describes how different components of the proposed system work together to solve the identified problem. For this project, the architecture provides a structured process for importing data from Google Sheets or CSV files into the target system using Import Sets and Transform Maps.

The main goals of the solution architecture are to:

- Provide an easy method to import data from Google Sheets or CSV files.
- Reduce manual data entry and the time required for importing large amounts of data.
- Map source fields to the correct target fields using Transform Maps.
- Transform data into the required format before importing.
- Validate the data and identify missing or invalid values.
- Handle duplicate records and prevent unnecessary duplicate entries.
- Identify failed records and provide import status and error information.
- Store the successfully processed data in the target table or database.
- Provide a reliable and organized data-import process.

Solution Architecture Flow

User → Google Sheets / CSV → Data Upload → Import Set Table → Transform Map → Field Mapping & Transformation → Data Validation → Duplicate Checking → Target Table / Database → Import Status & Error Logs

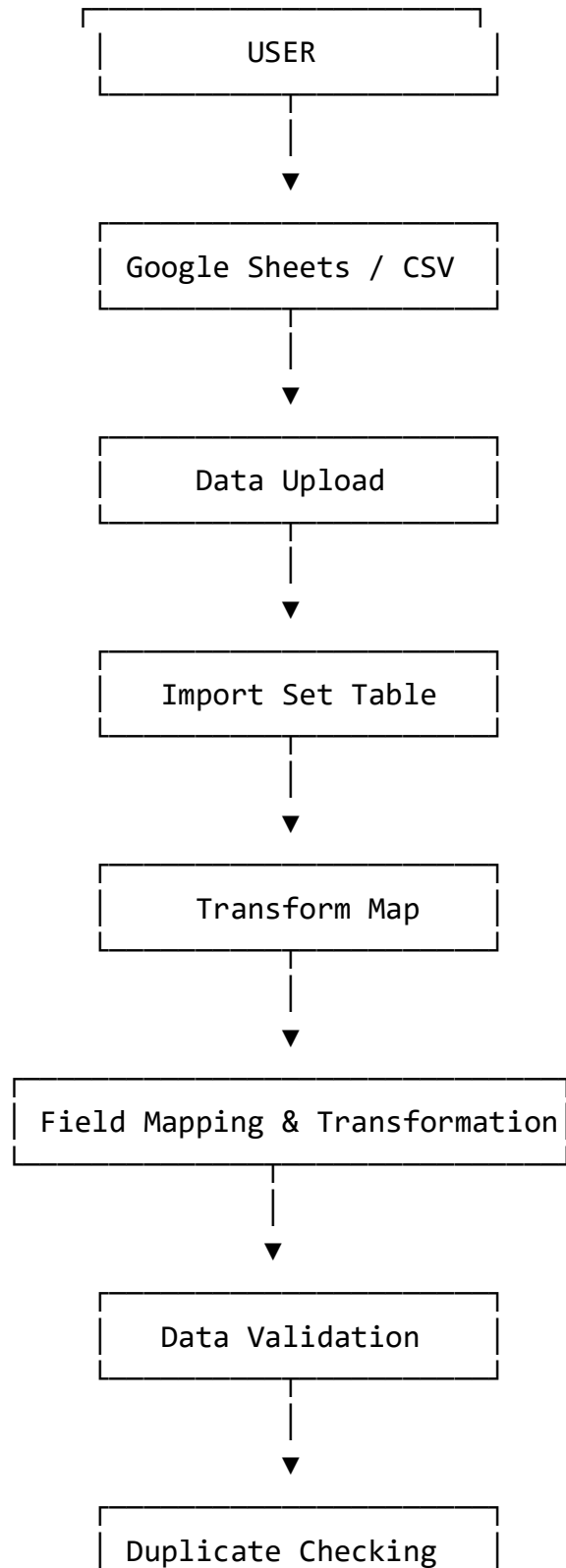
Architecture Components

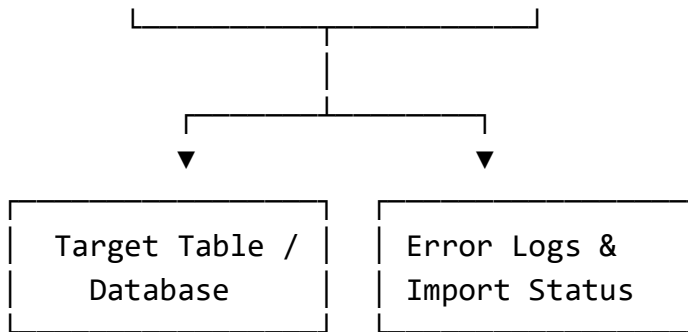
Component	Description
User	Uploads the data file and manages the import process.
Google Sheets / CSV	Acts as the source of the data to be imported.
Data Upload	Receives the source file and starts the import process.
Import Set	Temporarily stores the uploaded data before transformation.
Transform Map	Defines how source data should be transferred to the target table.
Field Mapping	Maps each source field to its corresponding target field.
Transformation Rules	Converts or modifies data into the required format.
Data Validation	Checks required fields, formats, missing values, and invalid records.
Duplicate Checking	Identifies existing or duplicate records before inserting data.
Target Table / Database	Stores the successfully imported and transformed records.
Error Logs	Stores information about failed or invalid records.
Import Status	Shows the success, failure, and overall result of the import process.

Working of the Architecture

1. The user prepares the required data in **Google Sheets or a CSV file**.
2. The file is uploaded into the system through the import process.
3. The uploaded data is temporarily stored in an **Import Set Table**.
4. A **Transform Map** is used to connect the source fields with the target fields.
5. Transformation rules are applied when the source data requires format or value changes.
6. The system validates the data and identifies missing or invalid records.
7. Duplicate records are checked to avoid unnecessary duplicate entries.
8. Valid and correctly transformed records are transferred to the **Target Table/Database**.
9. Failed records and errors are recorded in the import logs.
10. The user can check the **Import Status** and correct or reprocess failed records.

Architecture Diagram





Expected Outcome

The proposed architecture provides a systematic way to transfer data from Sheets or CSV files into the target system. It reduces manual effort, improves data accuracy, supports validation and duplicate handling, and makes it easier to monitor successful and failed imports.